Table 15.4 Radiation weighting factors, w_R (ICRP 2007)

LET in water (eV/nm)	Weighting factor w_R	Type and energy of radiation
0.2–35	1	photons (X-rays and γ's)
	1	electrons, muons
	2	protons and charged pions
130	20	α-particles, fission fragments, heavy ions
		continuous function of n-energy neutrons

biological effectiveness", RBE. The ICRP 1990 dose concept, the biologically effective dose, is termed *equivalent dose* and abbreviated H_T, and defined as:

$$H_T\,(\mathrm{Sv}) = \sum_R w_R D_{T,R} \tag{15.1}$$

$D_{T,R}$ is the absorbed dose averaged over the tissue organ T, due to radiation R. H_T is measured in sieverts (Sv); 1 Sv = 1 J kg^{-1}. The earlier, but similar concept was the *rem* (radiation equivalent man), where 1 Sv equaled 100 rem. Until 1990, H_T was called dose quivalent. $D_{T,R}$ is measured in J/kg, which in this context is called gray (Gy). The summation is taken over all sources irradiating the target.

15.3. RADIATION EFFECTS ON DIFFERENT TYPES OF CELLS

We can distinguish between two types of cells: those which are directly involved in the functioning of the organ (e.g. the cells of bone marrow, liver, or the nervous system) and those which are associated with reproduction. Radiation damage gives rise in the former to *somatic effects* (i.e. limited to the organism irradiated) such as cancer induction or cell death, and in the latter to *genetic effects* (i.e. limited to future generations).

Cells which are undergoing frequent division, and organs and tissues in which the cells are replaced slowly, exhibit high radiation sensitivity. Of the some 200 different kinds of cells in our body, some never divide (e.g. in the ovary, some sense organs and part of the central nervous system, except in the embryo state), while others divide frequently (bone marrow, intestinal epithelium, male gonads). The cell cycle time varies from hours to days. Usually, tumor cells divide much faster (3–5 times) than surrounding healthy tissue.

In general, the more differentiated the cells of an organ (i.e. the higher the organ is on the biological evolutionary scale), the greater the sensitivity to radiation. This also holds for different organisms, as reflected in Figure 15.2 and Table 15.5. Figure 15.2 shows the *inactivation dose* (i.e. leading to no further cell division) for organisms with different cell

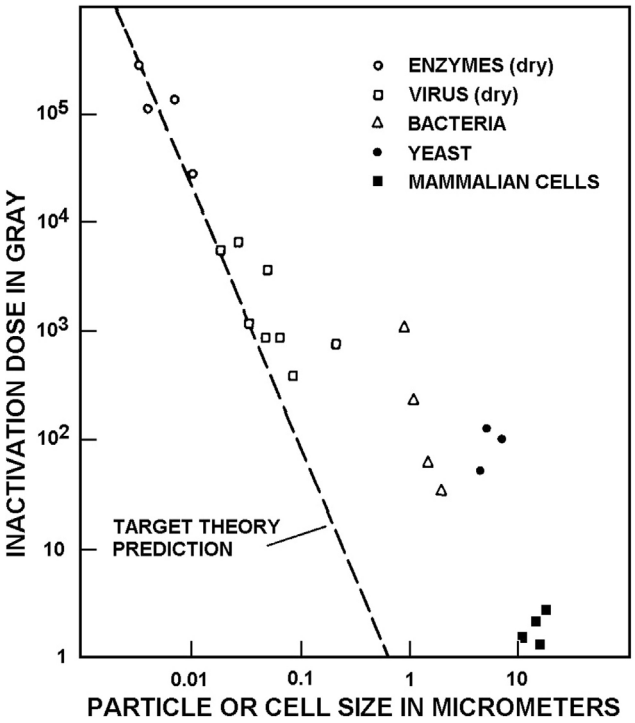


Figure 15.2 Inactivation doses for cells and particles of different sizes.

Table 15.5 Effects of γ-radiation doses on micro organisms, plants and animals

Doses for inactivation (Gy):	enzymes		> 20000
	virus (dry)		300–5000
	bacteria		20–1000
	human cells		≥1
Flowers (Senecio) survive at	10	Gy/d	during the
Trees do not survive at	>1	Gy/d	growing season
Trees normally survive at	≤ 0.02	Gy/d	(normally spring)
$LD_{50/30}$ (Gy) for	amoeba		1000
	fruit fly (Drosophila)		≥600
	shellfish		200
	goldfish		20
	tortoise		15
	song sparrow		8
	rabbit		8
	monkey		6
	man		~4
	dog		3.5

Table 15.6 Tissue weighing factors, w_T (ICRP 2007)

Body part	w_T	$\Sigma\, w_T$
Bone-marrow (red), colon, lung, stomach, breast, remainder	0.12	0.72
Gonads	0.08	0.08
Bladder, oesophagus, liver, thyroid	0.04	0.16
Bone surface, brain, salivary glands, skin	0.01	0.04
Total		1.00

$\Sigma\, w_T$ = sum of w_T-values. Remainder = Adrenals, extrathoracic region, gall, bladder, heart, kidneys, lymphatic nodes, muscle, oral mucosa, pancreas, prostate, small intestine, spleen, thymus, uterus/cervix

sizes. Table 15.5 gives doses which are inactivating or lethal to organisms within 30 days ($LD_{50/30}$) after short-time irradiation. A 10-fold dose is required to kill rather than to inactivate microorganism cells.

Because of the varying radiation sensitivities of the different cell types, a *tissue weighing factor*, w_T, is introduced. It represents the relative contribution of that tissue to the total detriment from uniform irradiation of the body; Table 15.6 gives tissue weighing factors. This leads to another dose concept (cf. eqn. 15.1), the *effective equivalent dose*, H_E, which is the sum of the weighted equivalent doses in all tissues, as given by

$$H_E\,(\mathrm{Sv}) = \sum_T w_T H_T \tag{15.2}$$

The weighing factors are important in medical radiation therapy and e.g. for evaluating the effects of internal radionuclides, §15.13.5. For low-LET whole-body irradiation w_R and $\Sigma w_T = 1$, i.e. the indici in (15.1) and (15.2) may be dropped.

15.4. SOME CONCEPTS OF RADIATION BIOLOGY

In radiology, the biological effects of radiation are usually discussed along two lines: the matrix effect and target theory. The *matrix effect* considers the particle-water interaction in which ions, radicals, and excited atoms are produced. This is the dominating effect at large radiation doses and dose rates; the lethal dose of 10 Gy generates about 10 000 electron tracks in each human cell. Free radicals and oxidizing products interact directly with cell DNA, causing the DNA-strands to break as described in §15.2.2. One can state that at such high doses the cell is simply poisoned by decomposition products, and the whole organ may be destroyed. It should be noted that in cancer therapy even much higher doses are given.

Inactivation of cell populations (Figure 15.2) and organisms (Table 15.5) requires comparatively large doses and the matrix effect should dominate. However, a 10 Gy dose to cells 1/1000 the size of human cells produces only 10 electron tracks per cell. The cell repair mechanisms are able to overcome the poisonous products. Consequently, doses